Research Article

Induced Paths, Radius, and Extremal Neighborhood Independence in Graphs without 4-Cycles

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Abstract

Let $M(n)$ be the maximum of $\ell(G) = S(G)/n$ over connected graphs of order n with no 4-cycle subgraph and no induced path on six vertices, where $S(G)$ is the sum of $\alpha(G[N(v)])$. This maximum is $16/7$, $5/2$, $8/3$ and 3 at orders 7, 8, 9 and 10, and $2+2/n$ for every $n \geq 11$; for every $n \geq 7$ the extremal graph is unique, and $S \leq 3n$ with equality only for the Petersen graph. A connected graph with no 4-cycle subgraph has radius 4 and induced-path order 7 if and only if it is C_8, or it is obtained from a 6-cycle by assigning to each of the six cycle vertices a nonempty set of outer vertices adjacent to that cycle vertex and to no other vertex of the cycle, each such set inducing a matching together with isolated vertices, and by placing at most one further vertex on each cycle edge; every such graph satisfies $S = 2n-b-m$. For connected triangle-free graphs with no 4-cycle subgraph and no induced path on seven vertices, $2e \leq 3n$ with equality only for the Petersen graph, and the maximum of $2e$ on $n \geq 10$ vertices is $3n - \min\{n-10,\ 5+\mathbf{1}_{\{n \text{ even}\}}\}$; the extremal graphs are determined, and the extremal family changes at the critical orders 15 and 16. Every connected graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $S \leq 3n$, with equality only for the Petersen graph. Consequently every connected graph with $\ell \leq 4$ satisfies Conjecture 133 of *Written on the Wall II*. The conjecture remains open for $\ell > 4$.

1 | Introduction

All graphs are finite and simple. For a graph G with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where α is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if G contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong C_4-free*.

The main results are Theorems 3.6, 4.3, 5.6 and 5.9. Let P denote the Petersen graph and $P - v$ the graph obtained by deleting one vertex. Theorem 3.6 determines the maximum $M(n)$ of ℓ over connected strong C_4-free P_6-free graphs of order $n \geq 7$. The value is $16/7$ at order 7, $5/2$ at order 8, attained only by the graph H_8 of Figure 1, $8/3$ at order 9, attained only by $P - v$, and 3 at order 10, attained only by P. For every $n \geq 11$ the maximum is $2 + 2/n$. For every $n \geq 7$, the extremal graph is unique up to isomorphism. The same theorem gives $S \leq 3n$, with equality only for P.

The shape of the extremal graph changes between orders 10 and 11, and the change is forced by the induced 6-cycle branch. A vertex off such a cycle meets it in an antipodal pair. Distinct off-cycle vertices of the first distance layer use distinct pairs, so there are at most three of them. If at least two antipodal pairs are occupied, any two vertices of the second distance

layer would have two common neighbours in the first layer. That layer therefore has size at most one, and the order is $6+k+h \le 10$. The graph of order 10 in this branch is P. At order 11 the branch with two or more occupied pairs is empty. Every remaining graph in the induced 6-cycle branch has a single off-cycle vertex a joined to one antipodal pair, and the second distance layer induces a matching together with isolated vertices. If that matching has m edges, the count in the proof of Theorem 3.6 gives $S = 2n + 2 - m$. Equality $S = 2n + 2$ holds only for $m = 0$, that is, only when every vertex of the second layer is a leaf, and the extremal graph is then the graph Γ_{n-7} of Figure 1. The class itself is not a single graph: the graph obtained from Γ_4 by joining two of its leaves is strong C_4-free and P_6-free of order 11, and it has $S = 23$, strictly below the extremal value 24. Every graph in the induced 5-cycle branch has $S \le 2n + 1$, and every chordal graph in the class has $S \le 2n - 2$, so both fall short of $2n + 2$. Uniqueness of the extremal graph can fail for smaller orders. The same case division shows that at order 4 both P_4 and $K_{1,3}$ attain $S = 6$, and that at order 6 both C_6 and the graph obtained from C_5 by adding one leaf attain $S = 12$.

An enumeration of the connected strong C_4-free P_6-free graphs of order at most 12 agrees with this uniqueness statement. It is not used in the proofs.

n	7	8	9	10	11	12
number of graphs	43	87	183	355	699	1327
maximum of S	16	20	24	30	24	26
number of extremal graphs	1	1	1	1	1	1

Theorem 4.3 describes every connected strong C_4-free graph of radius 4 and induced-path order 7. Fajtlowicz characterised the radius-critical graphs, and DeLaViña, Larson, Pepper and Waller record the consequence used here [10, 8]. An r-ciliate is a graph $C(2t, r-t)$ for some $1 \le t \le r$: one endpoint of each of $2t$ paths of order $r-t+1$ is joined in a cycle, with the extreme case $t = 1$ read as the path P_{2r} and the extreme case $t = r$ read as C_{2r}. The four 4-ciliates are therefore P_8, the 4-cycle with a pendant path of length 2 at each vertex, the 6-cycle with a pendant edge at each vertex, and C_8. The path P_8 is excluded by the hypothesis $p = 7$: that path has induced-path order 8, and it has no 4-cycle core. The 4-cycle with pendant paths of length 2 is excluded because the core is a 4-cycle subgraph. The two skeletons that remain are C_8 and the 6-cycle with a pendant leaf at each vertex. The new content is the classification of every extension of that skeleton. All six sets C_i are nonempty. Each C_i consists of outer vertices adjacent to the cycle vertex v_i and to no other vertex of the cycle, and $G[C_i]$ is a matching together with isolated vertices. These six sets may be nonempty simultaneously. The only further vertices are placed on cycle edges, at most one on each edge. The count is $S = 2n-b-m$, where b is the number of vertices added on cycle edges and m is the number of matching edges. Distance-4 witnesses are the device that forces this description. For each i the radius supplies a vertex $w_i \in C_i$ at distance four from v_{i+3}, and an edge with ends in distinct sets C_i and C_j completes an induced P_8 through two of these witnesses. Figure 2(a) draws that path. The same figure draws an isolated leaf, a matching edge and a glued vertex, so that the contributions to $b + m$ are visible. Corollaries 4.4 and 4.5 give S and the equality case. Neither theorem is a special case of the conjecture recorded next.

Conjecture 1.1 (Graffiti/WOWII Conjecture 133)**.** *Every connected graph G on at least two vertices satisfies*

$$r(G) + \ell(G)^{c_{C_4}(G)} \le p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [7]. We follow the formal statement in the Formal Conjectures repository at the fixed commit 5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e [11], the merge commit of pull request #5727 [12]. That commit removed the floor function from an earlier version of the file. The statement there reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. We work with this real-valued statement and call it REAL-133.

If G contains a 4-cycle, the conjecture reads $p(G) \ge r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \ge r(G)$ is induced, so the inequality holds. Only strong C_4-free graphs remain. Since p and r are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \;\ge\; \lceil \ell(G) \rceil. \tag{1.1}$$

Theorem 5.18 is the range in which the estimates that precede the full mass bound already give REAL-133. The triangle-free bound is Theorem 5.4, and Theorem 5.6 is the exact value of that maximum for every order $n \ge 10$, together with the isomorphism type of every graph that attains it. Theorem 5.9 extends $S \le 3n$, with equality only for the Petersen graph, from the triangle-free graphs to every connected strong C_4-free P_7-free graph. A minimal graph in the class that contains a triangle and satisfies $S \ge 3n$ has no cut vertex, satisfies $L(v) \ge 4$ at every vertex and has no two adjacent vertices of degree two, and Lemma 5.8 identifies the only graph with those local properties as an eleven-vertex subdivision of K_5 with $S = 29$. Corollary 5.19 therefore gives REAL-133 for every connected graph with $\ell \le 4$. The conjecture remains open for $\ell > 4$. Proposition 5.16 retains the bound $3 < \ell \le 4 - 16/n$ on the cell $r = 3$, $p = 6$, $T \ge 2$ and $n \ge 17$ as an intermediate estimate of the edge-deficit reduction; that cell is empty.

H. Chen [3] studies the floor form $p \geq r + \lfloor \ell \rfloor$ of the same conjecture. The arguments below do not use his proofs. His Lemma 2.1 is the local matching lemma: in a strong C_4-free graph the neighbourhood of every vertex is a disjoint union of edges and isolated vertices, and $S = 2e - 3T$. Lemma 2.1 records the same description. His Lemma 7.18 states that if an induced 6-cycle lies in a strong C_4-free P_6-free graph, then every vertex outside the cycle has either no neighbour on the cycle or exactly two, and those two are antipodal. His Corollary 7.20 records the consequence for the cycle vertices: each has triangle count zero, and its degree is 2 or 3 according as its antipodal slot is empty or occupied. His Lemma 7.22 applies to a graph in the class RES of his Definition 7.2 — a connected C_4-free graph of radius 2, diameter 3 and $\ell > 3$ in which every vertex of neighbourhood independence at least 3 has eccentricity at most 2 — that in addition has no induced P_6. In the forced hexagon of that argument, if P is an outer vertex and W_P is the set of second-layer vertices adjacent to P, part (b) states that every vertex of W_P is adjacent to the other outer vertex, and part (c) deduces from the absence of a 4-cycle that W_P contains at most one vertex. The same restriction on traces is Lemma 3.4 when the cycle dominates, and it is the description of the first distance layer in the proof of Theorem 3.6. His Theorem 6.7 states that a connected C_4-subgraph-free graph of diameter 2 with $\ell > 3$ has an induced path on at least six vertices. On the slice $r = 2$ and $3 < \ell \leq 4$ this is the same numerical conclusion as Theorem 5.18. The increment here is that Theorem 5.18 also covers radius 2 when the diameter is larger than 2, while Theorem 3.6 determines the extremal value at every order. Immediately after the proof of his Theorem 3.6, Chen records that every regular graph of girth at least 5 and radius 2 satisfies the floor form, and that this includes every Moore graph of diameter 2. A Moore graph of diameter 2 has ℓ equal to its degree, an integer, so the floor form and the real form coincide. The Moore assertion of Proposition S2.4 in the supplement is an alternate proof of that known case.

The exact function $M(n)$ is not a consequence of those lemmas. Lemma 7.22 controls the second layer only inside RES, and only after the hexagon has been forced. The proof of Theorem 3.6 removes those hypotheses. It treats the full class of connected strong C_4-free P_6-free graphs, continues to the second distance layer, and handles the induced 5-cycle branch and the chordal branch separately. For every $n \geq 7$ it also gives uniqueness. Below that range the same division produces two extremal graphs at orders 4 and 6, as recorded above. Theorem 5.4 is the two-step argument from the classification, due to Chen and Xu [5, Lemma 3.4], of the connected triangle-free strong C_4-free P_7-free graphs with no clique cutset. If v is a cut vertex, induced paths that start at v in distinct components of $G - v$ concatenate, so one side has rooted path order at most 3 and falls under Lemma 5.3. If there is no cut vertex but $\{u, v\}$ is a clique cutset, a shortest u–v path in each of two components has three internal vertices, and those six vertices together with v induce a P_7. Restoring clique cutsets therefore yields no new equality graph: every graph with a clique cutset is strictly below $2e = 3n$, and the only equality graph on the Chen–Xu list is the Petersen graph.

Examples S3.1 and S3.7 in the supplement start from an r-ciliate of larger radius and add further blocks. They are not part of the radius-4 classification. In Example S3.1 each added path of length four contributes 3 vertices and 4 edges, so it increases $S = 2e$ by 8 while the radius and the induced-path order remain those of the ciliate. In Example S3.7 each attached copy of the incidence graph of a projective plane adds $(q + 1)f$ to S and $f - 1$ to the order, and the radius and the path order remain $t + 3$ and $2t + 5$.

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves the triangle-free bound, the exact function $f(n)$ and the extremal graphs, the one-triangle bound, and Theorem 5.9. Section 6 is the application to REAL-133 for $\ell \leq 4$, including the θ_k examples, and it records that the range $\ell > 4$ remains open. The rooted inequalities, the endpoint obstructions, the high-radius constructions, the finite-search certificate for Lemma 5.8, and the candidate pairs in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$ are in the supplement. The conditional reduction formerly placed there is superseded by Theorem 5.9. Those supplement arguments are not used for Theorems 3.6, 4.3, 5.4 and 5.6. Table 1 is the list of what is proved and what is open. A finite search is a proof in this article only together with a completeness argument in the text and a versioned checker in the submission. Lemma 5.8 is of that form. An enumeration that does not meet this standard is not used as a hypothesis.

2 | Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing v. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

Lemma 2.1. *Let G be strong C_4-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every v, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

Proof. (1) Two common neighbours w, w' of x, y give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with v. So the components are single vertices and edges, each edge being a triangle at v. (3) Sum $d(v) - t(v)$ over v. □

The components of $G[N(v)]$ are the *units* of v. For a vertex set U, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of U. We write $L(v) = D_G(\{v\})$.

Lemma 2.2. *Let G be strong C_4-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.

Proof. For $x \notin U$, deleting U removes exactly the units of x that lie inside U; a unit that is an edge with one end in U survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of v in triangles with v keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

Lemma 2.3. *Let v be a cut vertex of G. Split the components of $G - v$ into two nonempty groups, and let G_1, G_2 be the subgraphs induced by the two groups together with v. Then $S(G) = S(G_1) + S(G_2)$.*

Proof. $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

Theorem 2.4 (Erdős, Saks and Sós [9])**.** *Every connected graph G has an induced path on at least $2r(G) - 1$ vertices.*

Theorem 2.5 (Camby and Schaudt [2, Theorem 5])**.** *Let $k \geq 4$ and let G be a connected graph with no induced path on k vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle C_k.*

Theorem 2.6 (Seinsche [14])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [2, Theorem 5] applied with $H = G$. We write P_k for the path on k vertices, and "G is P_k-free" means that G has no induced P_k, that is, $p(G) \leq k - 1$. A connected graph of radius r has at least $2r$ vertices.

Lemma 2.7. *Let H be a connected strong C_4-free P_4-free graph on at least two vertices. Then H has a vertex adjacent to all other vertices.*

Proof. By Theorem 2.6, H is the join of two nonempty vertex sets A and B. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

3 | Small-path mass bounds

Lemma 3.1. *If G is connected and strong C_4-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where m is the number of edges of $G[N(u)]$ for a dominating vertex u. Hence $\ell(G) < 2$, and (1.1) holds.*

Proof. Let $v \neq u$. By Lemma 2.1(1), v has at most one neighbour other than u, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \leq 2$. If G is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with u form an induced P_3, so $p \geq 3 \geq r + \lceil \ell \rceil$. □

Lemma 3.2. *If G is connected and strong C_4-free with $r(G) \geq 2$, then $p(G) \geq 4$.*

Proof. If G were P_4-free, Lemma 2.7 would give a dominating vertex, so $r(G) \leq 1$. □

Lemma 3.3. *If G is connected, strong C_4-free and P_5-free, then $S(G) \leq 2n - h$, where h is the number of vertices v with $\lambda(v) \geq 3$. In particular $S(G) \leq 2n$.*

Proof. Call v *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \geq 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of x if x lies in an edge unit of v.

Key fact. Let x, y lie in different units of v. A common vertex of X_x and X_y would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced P_5: $x \nsim y$ because they lie in different units, $x \nsim y'$ and $y \nsim x'$ by disjointness, and v is adjacent to neither x' nor y'. So X_x is complete to X_y. If both are nonempty and $|X_x| \geq 2$, a vertex of X_y has two neighbours in X_x, which are also neighbours of x, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.

Main step. For each high v we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is v, one in each of $\lambda(v) - 1$ distinct units of v.

Case 1: some $u \in N(v)$ has $|X_u| \geq 2$. By the key fact $X_y = \emptyset$ for every y outside the unit of u. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so y is simplicial; so is $\bar{y}$. Hence v is the only high neighbour of y, and the $\lambda(v) - 1$ units other than that of u supply the vertices.

Case 2: $|X_x| \leq 1$ for all $x \in N(v)$. Then $\lambda(x) \leq 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices x has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let x lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced P_5: $x \nsim y$ (different units), $x \nsim y', z'$ (disjointness), $v \nsim y', z'$, and $y \nsim z'$ (disjointness). As $\lambda(v) \geq 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is v. This gives at least $\lambda(v) - 1$ units.

Counting. The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}} (\lambda(v) - 1) \leq \#\{\text{simplicial vertices}\}$, and

$$S = \sum_{v} \lambda(v) \leq 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \leq 2n - h. \qquad \square$$

Lemma 3.4. *Let G be connected, strong C_4-free and P_6-free with a dominating induced cycle $C = v_0v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of C, distinct vertices of W use distinct pairs, and W is independent. Hence $|W| \leq 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \leq 2$ if $W \neq \emptyset$.*

Proof. Indices are modulo 6. Let $w \in W$; it has a neighbour on C. If w has exactly one neighbour v_i, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced P_6. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced P_6. Two neighbours at distance 2 on C give a 4-cycle. A set of at least three vertices of C_6 contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of W with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on C, and wv_iv_jw' is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if w uses $\{v_i, v_{i+3}\}$, then every vertex of C is within distance 2 of v_i (the vertex v_{i+3} through w), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So v_i has eccentricity 2. $\square$

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong C_4-free P_6-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form.

Lemma 3.5. *Let G be connected, strong C_4-free and P_6-free. Then either G has a vertex of eccentricity at most 2, or $G = C_6$.*

Proof. Apply Theorem 2.5 with $k = 6$ to get a connected dominating set X. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is P_4-free. If $|X| = 1$, the vertex in X dominates G. If $|X| \geq 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since X dominates G, every vertex is within distance 2 of u. $\square$

Theorem 3.6. *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong C_4-free P_6-free graphs of order $n \geq 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \geq 11. \end{cases}$$

For every $n \geq 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph P. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph H_8 of Figure 1. For $n = 7$ and for every $n \geq 11$ it is the graph Γ_{n-7} of that figure: a 6-cycle, a vertex a adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to a. In particular, every connected strong C_4-free P_6-free graph G of order $n \geq 2$ satisfies $S(G) \leq 3n$, with equality if and only if G is the Petersen graph. At orders 4 and 6 the maximum is attained by two nonisomorphic graphs.

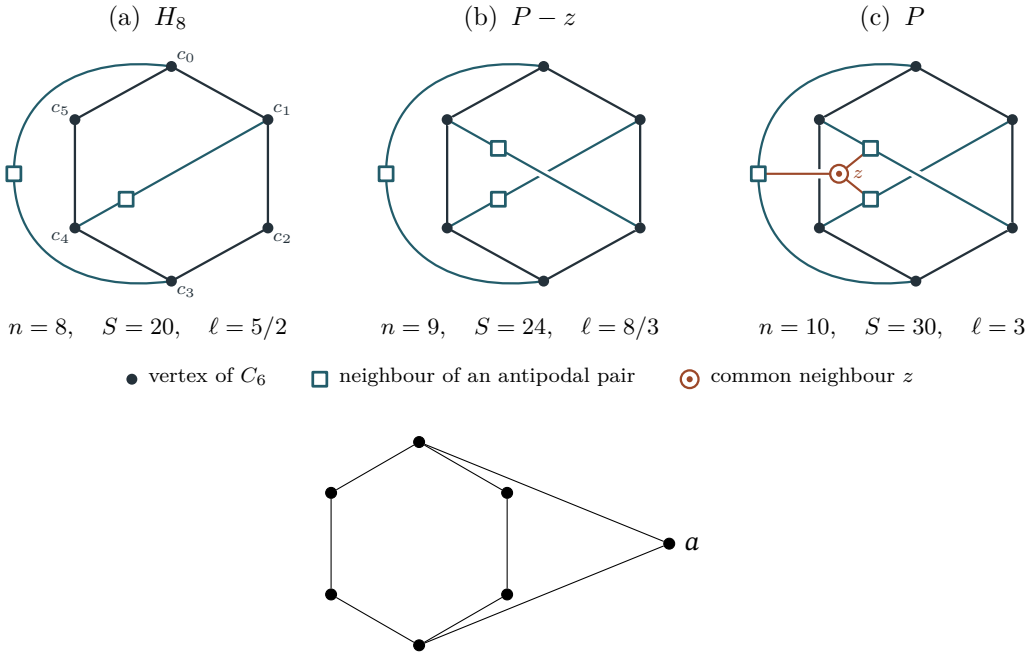


FIGURE 1 | Extremal graphs for $M(n)$. In H_8, two antipodal pairs on the 6-cycle have a common neighbour outside the cycle. Adding a common neighbour for the third pair gives $P - z$. Joining a new vertex z to the three outside vertices gives the Petersen graph P. The three graphs are triangle-free, so $S = 2e$. The drawing underneath is Γ_0, the unique graph of order 7 with $\ell = 16/7$: the marked vertex a is adjacent to one antipodal pair. Attaching any number of leaves at a gives the unique extremal graph of each order $n \geq 11$, with $S = 2n + 2$.

Proof. Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced P_6. Thus the only possible holes have lengths five and six.

Case 1: an induced six-cycle. Label its vertices $c_0, \ldots, c_5$, indices modulo 6, and write A for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of A use distinct pairs, and A is an independent set: an edge of A joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \leq 3$. Figure 1 shows the three extremal graphs of this branch.

Every vertex x at distance two from the cycle is adjacent to every vertex of A. Such an x has at least one neighbour in A. If $x \sim a$ and $x \nsim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of b lies at cycle distance two from some contact of a, and we orient the cycle from that contact of a toward that contact of b. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced P_6. Indeed x meets the cycle in nothing, a meets the cycle only in its antipodal pair (neither c_{i+1} nor c_{i+2}), b meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \nsim c_i, c_{i+1}$), and $a \nsim b$.

There is no vertex at distance three or more from the cycle. A shortest path z, x, a, c_i from such a vertex down to the cycle, extended by c_{i+1}, c_{i+2}, is an induced P_6: the vertex z has no neighbour in A or on the cycle, and x has no neighbour on the cycle.

Let h be the number of vertices at distance two. If $k = 0$, then A is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with a. Writing m for the number of edges of the matching gives $n = 7+h$, $e = 8+h+m$ and $T = m$, so $S = 16+2h-m = 2n+2-m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that G is the graph Γ_h formed by the 6-cycle, the vertex a on one antipodal pair, and h leaves at a.

The graph Γ_h lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from a to the cycle, and each of the two a-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced P_6. A leaf, having degree one, occupies an end of any path that contains it, and that path meets a next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in Γ_0 that starts at a. No such path exists: the two neighbours of a on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord ac_{i+3}, and every shorter continuation from a through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to a. An induced path of Γ_0 that avoids a lies on the 6-cycle and has order at most five. An induced path of Γ_0

that contains a in its interior has c_i and c_{i+3} as the two neighbours of a on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus Γ_h is strong C_4-free and P_6-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of A. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of A and to nothing on the cycle, A is independent, and each vertex of A contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

(k, h)	$(2, 0)$	$(2, 1)$	$(3, 0)$	$(3, 1)$
S/n	$5/2$	$8/3$	$8/3$	3

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write a_i for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices a_0, a_1, a_2 and a vertex z adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding a_i; the vertex z meets every a_i; and a non-neighbour of a_i on the cycle is adjacent to c_i or to c_{i+3}. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives v_0 a common neighbour with each of v_3, v_4, v_5. Those neighbours lie off the path, because the only neighbour of v_0 on the path is v_1 and the path is induced. They are distinct: a vertex adjacent to two of v_3, v_4, v_5 would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: v_0 has degree 3, so it has only two neighbours off the path. Thus the graph is P_6-free. It is the Petersen graph P, and $\ell(P) = 3$.

Deleting z leaves $(k, h) = (3, 0)$, the graph $P-z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex z, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by c_5, z and c_2, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of P are strong C_4-free and P_6-free. The pair $(2, 0)$ is the graph H_8 in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither P nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.

Case 2: an induced five-cycle and no induced six-cycle. Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from C has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on C and forms a 4-cycle. If a vertex x were at distance two, with a neighbour a in the first layer, the path x, a followed by four consecutive cycle vertices, beginning at one contact of a and omitting the second contact if a has one, would be an induced P_6. Thus C dominates G. Write h for the number of vertices with singleton trace and j for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside C are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of C. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let m be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at c_i meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of c_i and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let b be the number of singleton-to-pair edges, so $b \leq \min(h, j)$.

Consequently $n = 5+h+j$ and $e = 5+h+2j+m+b$. The triangles are the j pair triangles and the m matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer b is at most 1. Two singleton–pair edges would use two distinct supports: the opposite edge of c_i is $\{c_{i+2}, c_{i+3}\}$, these edges exhaust the cycle as i varies, and each pair class contains at most one vertex. The graph is (P_6, C_4)-free and, in this case, has no induced C_6, so Theorem 3.2(d) of Karthick and Maffray applies [13]. If the two supports lie at distance 2 on C, those two singleton classes cannot both be nonempty. If they lie at distance 1, the two corresponding pair classes cannot both be nonempty. Either alternative contradicts the existence of both edges, so $b \leq 1$.

If $b = 0$, then $S = 2n - j - m \leq 2n$. If $b = 1$, then $j \geq 1$ because $b \leq \min(h, j)$, so $S = 2n + 2 - j - m \leq 2n + 1$. In this case $S \leq 2n + 1$.

Case 3: no holes. The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex

of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong C_4-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \leq 2(n-1) = 2n-2$.
The three cases give the values of $M(n)$. A graph of order $n \geq 11$ cannot realise $k \in \{2,3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs Γ_h and the graphs with the same first layer and $m \geq 1$. The latter have $S \leq 2n+1$. Cases 2 and 3 give $S \leq 2n+1$ and $S \leq 2n-2$. Thus $M(n) = 2 + 2/n$, attained only by Γ_{n-7}.
At order 10 the only graph in the list with $S > 2 \cdot 10 + 2$ is P, with $S = 30$. At order 9 the only such graph is $P - v$, with $S = 24$, whether it arises as $(k,h) = (3,0)$ or as $(k,h) = (2,1)$. At order 8 the only such graph is H_8, with $S = 20$. At order 7 one has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 16$ is Γ_0, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \geq 11$, the bound $S \leq 3n$ holds for every order $n \geq 2$, with equality only for P. The case division does not use $n \geq 7$. At order 4 there is no cycle of length five or six, so only the chordal bound applies and $S \leq 6$. Equality holds for a tree, and the only trees on four vertices are P_4 and $K_{1,3}$. At order 6 the induced 6-cycle branch contributes only C_6, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \leq 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph C_5 with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly two graphs. □

Remark 3.7. Karthick and Maffray describe the structure of graphs with no induced P_6 and no induced 4-cycle, and deduce the colouring bound $\chi \leq \lceil 5\omega/4 \rceil$ [13]. Their Theorems 3.2 and 3.3 describe the neighbourhood, on an induced C_5 or C_6, of a vertex outside the cycle. Specialising the 6-cycle description to graphs with no 4-cycle subgraph excludes a consecutive triple, a consecutive quadruple and the full cycle as a support, and leaves an antipodal pair. That is the trace statement at the beginning of Case 1. Part (d) of their Theorem 3.2 is the C_6-free restriction used to bound the number of singleton–pair edges in Case 2. The constraint on the second layer, the bound $h \leq 1$ when $k \in \{2,3\}$, and the exact function $M(n)$ are further arguments. They are not consequences of the local support list, and the colouring theorem is not a substitute for the extremal statement under the subgraph prohibition.

Remark 3.8. Brandstädt and Hoàng analysed (P_6, C_4)-free graphs through clique cutsets [1]. In that language C_4 is an induced cycle. A graph in the class with no clique cutset has every induced C_5 dominating, and the presence of an induced C_6 that fails to dominate forces the graph to be the join of a blow-up of the Petersen graph and a clique, possibly empty. Theorem 3.6 is a different statement. The prohibition here is that of a 4-cycle as a subgraph, and the conclusion is the exact maximum of S over every connected graph in the resulting class, including graphs that have clique cutsets. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither construction enters the class.

Remark 3.9. An alternative proof of $S \leq 3n$, with equality only for the Petersen graph, fixes a vertex of eccentricity at most 2 by Lemma 3.5 and charges the cross edges in the second distance layer. The case division above already gives that inequality and the equality case, so the charging argument is not repeated.

4 | Radius four

Lemma 4.1. *Let G be connected, strong C_4-free and P_8-free. If an induced 8-cycle dominates G and $r(G) \geq 4$, then $G = C_8$.*

Proof. Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex y the trace $A_y = N(y) \cap V(C)$ is nonempty. The trace A_y is not contained in any three consecutive vertices of C. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends c_i and c_{i+2} are not both in A_y, since y, c_i, c_{i+1}, c_{i+2} would be a 4-cycle. Thus A_y is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact c_{i+1} does not lie on the path. Either path is an induced P_8.
Fix a cycle vertex, say c_0. The vertices at cycle distance greater than two from c_0 are the three consecutive vertices c_3, c_4, c_5. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of c_0. The same holds with c_0 replaced by any vertex of C.
Suppose w lies outside C, and write $A = N(w) \cap V(C)$. No two vertices of A are at cycle distance two, for they would form a 4-cycle with w. A set of vertices of C in which every two distinct vertices are consecutive has size at most two, and

every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities have been excluded. Therefore A contains two vertices at cycle distance three or four. Rotating, take them to be c_0 and c_d with $d \in \{3, 4\}$.
The antipode of c_0 is c_4. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of C is already at cycle distance at most three from c_0. Every outside vertex is within distance three of c_0. Thus the eccentricity of c_0 is at most three, contradicting $r(G) \geq 4$.
Hence C has no vertex outside it, and $G = C_8$. □

Remark 4.2. The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced P_8. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced P_8. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order $8, \ldots, 13$ the counts are $1, 32, 262, 384, 107, 4$. It finds 73 isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

Theorem 4.3. *A connected graph G is strong C_4-free and satisfies $r(G) = 4$ and $p(G) = 7$ if and only if either $G \cong C_8$, or G is obtained from a 6-cycle $C = v_0 v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each i there is a nonempty set C_i of vertices outside C. Every vertex of C_i is adjacent to v_i and to no other vertex of C. There is no edge between C_i and C_j for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of C there is at most one further vertex, adjacent precisely to the two ends of that edge.*

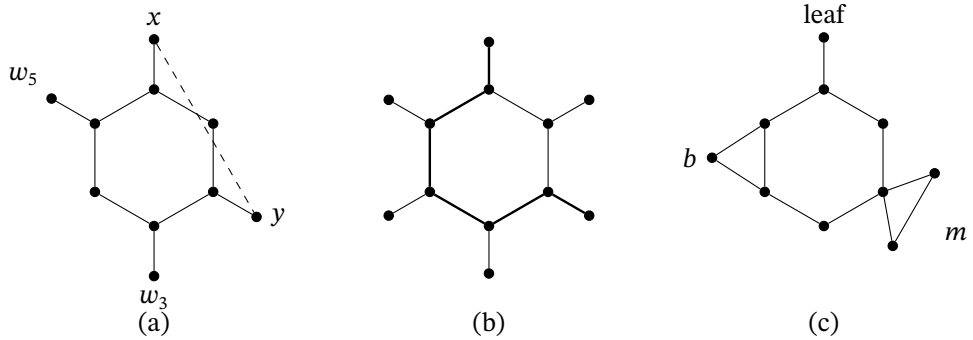


FIGURE 2 | The radius-four case of Theorem 4.3. (a) If x and y have singleton traces $\{v_0\}$ and $\{v_2\}$ on the dominating 6-cycle, the dashed edge xy completes the induced path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The vertices w_5 and w_3 are the distance-four witnesses of the proof. (b) One leaf at each cycle vertex gives $S = 2n$. The heavy path has seven vertices. (c) On the same cycle: an isolated leaf, a matching edge inside one set C_i, and a vertex glued to a cycle edge. The leaf does not contribute to $b + m$. The matching edge and the glued vertex each increase $2n - S = b + m$ by 1.

Proof. First suppose that G is strong C_4-free, with $r(G) = 4$ and $p(G) = 7$. Theorem 2.4 gives $p(G) \geq 7$, which agrees with the hypothesis. Apply Theorem 2.5 with $k = 8$. There is a connected dominating set X such that $G[X]$ is P_6-free or $G[X] \cong C_8$.
If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.
If $G[X]$ is P_6-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in G, because X dominates G, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.
Traces. As in the proof of Lemma 3.4, strong C_4-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some w has trace $\{v_i, v_{i+3}\}$, then v_i is within distance two of every vertex of C, hence within distance three of every vertex of G, a contradiction. So there are no antipodal traces. Let C_i be the set of vertices with trace $\{v_i\}$.
Witnesses. Fix i. Since $r(G) = 4$, some vertex is at distance at least four from v_{i+3}. Every vertex of C is within distance three of v_{i+3}, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from v_{i+3}, which is at most three unless the trace of w is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from v_{i+3}. Every neighbour $z \in W$ of w_i lies in C_i: if z had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if z had the neighbour v_{i-1} or v_{i+1}, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.
Edges in W. Figure 2 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm 1}$: $x, v_0, v_{\pm 1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced P_8.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced P_8. By symmetry, vertices of C_i and C_j with $i \neq j$ are never adjacent.
- x with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices v_0, v_1, x, y span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle x, v_0, v_5, y (respectively x, v_1, v_2, y) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced P_8, and $j = 4$ is symmetric.
- x, y with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm 1$ and $j = \pm 2$ give 4-cycles as above; $j = 3$ gives the induced P_8 $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside W joins two vertices of the same class C_i, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any C_j and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class C_i is nonempty. This is the stated construction.

Conversely, let G be C_8, or let G be built from $C = v_0 \cdots v_5$ as in the statement. The cycle C_8 has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on C. There is no edge between distinct sets C_i, and every vertex of C_i meets C only at v_i. A cycle that meets C_i is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at v_i together with a matching, so the only cycles are the triangles formed by v_i and one matching edge.

Let y be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through y uses both edges at y, and the rest of the cycle is a v_j–v_{j+1} path avoiding y. A 4-cycle through y is then a path of length two between v_j and v_{j+1} that avoids y: the two ends of the edge have a common neighbour other than y. No vertex of any C_i is adjacent to two vertices of C, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through y. The only cycle that uses no vertex outside C is C itself. Thus G has no 4-cycle subgraph.

Let P be an induced path. An isolated vertex of some $G[C_i]$ has degree one, so it is an end of P if it lies on P. A vertex of a matching edge in C_i has neighbourhood $\{v_i, x'\}$, a clique. It is therefore an end of P whenever P has a third vertex, and P cannot contain both ends of the matching edge together with v_i. A vertex added on a cycle edge has neighbourhood equal to that edge, again a clique, so it too is an end of any induced path that contains it, and the path cannot contain both ends of the cycle edge. Consequently every induced path of order at least three meets $V(G) \setminus V(C)$ in at most two vertices, both ends of the path. The remaining vertices lie on C, which has no induced path on six vertices. Hence $p(G) \leq 7$. Since each C_i is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: x and y meet the path only at v_0 and v_2, and C contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets C_i, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along C. The distance from v_i to v_{i+3} is therefore three. A vertex of C_{i+3} is at distance four from v_i, and the eccentricity of v_i is at least four. It is at most four: every vertex of C_j is at distance $d_C(v_i, v_j) + 1 \leq 4$ from v_i, and every vertex added on a cycle edge is within distance four of v_i. A vertex of C_i is at distance five from a vertex of C_{i+3}. A vertex y added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from v_j and from v_{j+1} to v_{j+3} are 3 and 2, neither neighbour of y is adjacent to v_{j+3}, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of C_{j+3} is at distance four from y. Every eccentricity is at least four, and $r(G) = 4$. □

Corollary 4.4. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let b be the number of vertices added on edges of the 6-cycle and let m be the number of matching edges in the sets C_i. Then $S(G) = 2n - b - m$. In particular $S(G) \leq 2n$, so $\ell(G) \leq 2$, and* RAD(4) *holds.*

Proof. For $G \cong C_8$ one has $n = 8, e = 8, T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, a edges from the sets C_i to the cycle, m matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

Corollary 4.5. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or G is obtained from C_6 by attaching at least one leaf to each cycle vertex.*

Proof. By Corollary 4.4, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each C_i is nonempty, so each cycle vertex has at least one leaf. The converse is the case

$b = m = 0$ of the construction, together with C_8, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

Remark 4.6. If G is connected and strong C_4-free with $r(G) = 4$, $p(G) \geq 8$ and $\ell(G) \leq 4$, then $p(G) \geq 8 = r(G) + 4 \geq r(G) + \ell(G)$. This comparison is not a structural description of G.

5 | Graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \dots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; F_5 is the Petersen graph, and the parameters t and t' attached to F_2 and F_0' are positive integers. The edge counts below use those constructions rather than the accompanying figure.

Lemma 5.1 (Chen and Xu [5, Lemma 3.4])**.** *Let G be a connected (P_7, C_4, C_3)-free graph with no clique cutset. If G is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- F_1 is C_5. The graph F_2 is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets A_1 and A_3 of equal positive order t, with a perfect matching between them, each vertex of A_1 adjacent only to v_1 and its match, and each vertex of A_3 adjacent only to v_3 and its match.
- On the same 5-cycle, F_3 has three further vertices inducing the path $a_3a_1a_4$, with a_i adjacent to v_i. The graph F_4 has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph F_5 is the Petersen graph.
- F_0 is C_6. The graph F_0' is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets Y_1 and Y_4 of equal positive order t', with a perfect matching between them, each vertex of Y_1 adjacent only to v_1 and its match, and each vertex of Y_4 adjacent only to v_4 and its match.
- F_6 is C_7. On a 7-cycle $v_1 \cdots v_7$, write x_i for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then F_7 is the cycle plus x_1; F_8 is the cycle plus nonadjacent x_1, x_2; F_9 is F_8 plus a vertex adjacent only to x_1 and x_2; F_{10} is the cycle plus nonadjacent x_1, x_5; F_{11} is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and F_{12} is F_{11} plus a vertex adjacent only to x_1 and x_2.

Chen and Xu record that F_0, F_1, the graph F_2 with $t = 1$, F_3 and F_4 are induced subgraphs of the Petersen graph, and that $F_6, \dots, F_{11}$ are induced subgraphs of F_{12}. The graphs $F_6, \dots, F_{12}$ are the extensions of C_7 by vertices of cycle-distance three. The graphs F_2 and F_0' are the matching extensions of one pair of opposite vertices on C_5 and on C_6.

Lemma 5.2. *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \leq 3n - 3$. The Petersen graph satisfies $2e = 3n$.*

Proof. The constructions give the following orders and sizes. For F_2 and F_0' the parameters t, t' are at least one.

Graph	n	e	$2e$	$3n - 2e$
$F_1 = C_5$	5	5	10	5
F_2	$5 + 2t$	$5 + 3t$	$10 + 6t$	5
F_3	8	10	20	4
F_4	9	12	24	3
F_5	10	15	30	0
$F_0 = C_6$	6	6	12	6
F_0'	$6 + 2t'$	$6 + 3t'$	$12 + 6t'$	6
$F_6 = C_7$	7	7	14	7
F_7	8	9	18	6
F_8, F_{10}	9	11	22	5
F_9, F_{11}	10	13	26	4
F_{12}	11	15	30	3

The five edges of F_3 beyond C_5 are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of F_4 beyond C_5 are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex x_i contributes two edges, and each vertex placed on a pair x_ix_j contributes two further edges. The Petersen graph is the only row with difference zero. □

Lemma 5.3. *Let F be connected and strong C_4-free, and let s be a vertex at which no induced P_4 starts. Write $m = e(F[N(s)])$. Then $S(F) \leq 2(|F| - 1) - m$.*

Proof. A vertex at distance 3 from s would end a geodesic, hence an induced P_4, starting at s. So $V = \{s\} \cup N_1 \cup N_2$ with N_i the i-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of y with a different parent would give the induced path $sayz$. So y has at most one neighbour in N_2, a sibling (two siblings would be common neighbours of a and y), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where C_a is the set of children of a and $\bar{a}$ is the neighbour of a in N_1, if any. Since s and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F| - 1) - m$. □

Theorem 5.4. *Let G be a connected triangle-free strong C_4-free P_7-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if G is the Petersen graph. In particular every connected triangle-free strong C_4-free graph of induced-path order at most 6 satisfies GEN_3, and so does every connected bipartite strong C_4-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

Proof. The proof is by induction on $n = |V(G)|$. The graph K_2 has $2e = 2 < 3n$. Suppose $n \ge 3$, and write $S = 2e$, which is legitimate because G is triangle-free.
Suppose first that v is a cut vertex. For each component C of $G - v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ that starts at v. Paths from two components form an induced path of G on $\rho_C + \rho_{C'} - 1 \le 6$ vertices, so some component C_1 has $\rho_{C_1} \le 3$. Let $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with orders n_1 and n_2. Then $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Lemma 2.3 gives $S(G) = S(G_1) + S(G_2)$. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$. The graph G_2 is a smaller connected triangle-free strong C_4-free P_7-free graph, so the inductive hypothesis gives $S(G_2) \le 3n_2$. Therefore

$$S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

If G has a cut vertex, this estimate gives the theorem, and equality does not hold. Assume for the remainder of the argument that G has no cut vertex.
Suppose that $\{u, v\}$ is a clique cutset of order two. Then uv is an edge. Since neither vertex is a cut vertex, every component of $G - \{u, v\}$ is adjacent to both u and v, and there are at least two such components. A shortest u–v path through one component has at least four edges: one internal vertex would form a triangle with uv, and two internal vertices would form a 4-cycle. In two components, choose shortest paths, and on each path let x, y, z be the three vertices immediately preceding v, in that order from the u-side. The same choice in the second component produces x', y', z'. The path x, y, z, v, z', y', x' is an induced P_7. Indeed, each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \le 6$.
Thus G has no clique cutset. A triangle-free graph has no 4-cycle subgraph if and only if it has no induced 4-cycle, so Lemma 5.1 applies: G is complete or belongs to $\mathfrak{F}$. The only complete triangle-free graph on at least two vertices is K_2, already strict. Lemma 5.2 gives $2e \le 3n$ for every graph in $\mathfrak{F}$, with equality only for the Petersen graph.
A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong C_4-free P_7-free graph.
Equality holds only for the Petersen graph. In particular, a graph with a clique cutset lies strictly below the bound, whether the cutset is a single vertex or an edge. □

Lemma 5.5. *Let G be a connected triangle-free strong C_4-free P_7-free graph of order n. Then G is obtained from a connected triangle-free strong C_4-free P_7-free graph Q with no clique cutset by attaching trees at single vertices. The core Q is K_1, K_2, or a graph in the list of Lemma 5.2. Writing $h = |V(Q)|$ and $\eta(Q) = 3h - 2e(Q)$, one has $e(G) = e(Q) + (n - h)$ and*

$$3n - 2e(G) = \eta(Q) + n - h.$$

Deleting the vertices of one attached tree, other than its root on Q, deletes exactly as many edges as vertices.

Proof. If G itself has no cut vertex, take $Q = G$. The edge-separator argument in the proof of Theorem 5.4 excludes a clique cutset of order two, and G is triangle-free, so it has no clique cutset. Lemma 5.1 places a non-complete core among the graphs of Lemma 5.2, and the only complete triangle-free graphs are K_1 and K_2. The identities are then immediate.
Suppose s is a cut vertex. As in Theorem 5.4, path concatenation supplies a component C of $G - s$ such that every induced path of $G[C \cup \{s\}]$ that starts at s has order at most three. The first distance layer of this side is independent, since an edge there would form a triangle with s. There is no vertex at distance three or more from s, for such a vertex would end an induced P_4 at s. A vertex of the second layer has a unique parent: a second parent, together with s, would span a 4-cycle. It has no neighbour in its layer. An edge joining vertices of distinct parents is the induced path of order four recorded in the proof of Lemma 5.3, and an edge joining two children of the same parent forms a triangle with that parent. Thus $G[C \cup \{s\}]$ is a tree. Deleting the $k = |C|$ vertices of the side deletes exactly k edges. The graph that remains is connected, triangle-free, strong C_4-free and P_7-free. Iterate. The remaining graph Q has no cut vertex, and the first paragraph describes it. Reversing

the deletions attaches trees at single vertices. Each deleted vertex accounted for one edge, so $e(G) = e(Q) + (n - h)$, and the displayed identity follows. □

Theorem 5.6. *Let $f(n)$ be the maximum of $2e(G)$ over connected triangle-free strong C_4-free P_7-free graphs of order n. For every $n \geq 10$,*

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

Write $\mathcal{P}(n)$ for the isomorphism classes of graphs obtained from the Petersen graph by placing $n - 10$ leaves on a single vertex or on the two endpoints of a single edge. For $10 \leq n \leq 14$ the extremal graphs are exactly the classes in $\mathcal{P}(n)$. For $n = 15$ they are the classes in $\mathcal{P}(15)$ together with F_2, four classes in all. For $n = 16$ they are the classes in $\mathcal{P}(16)$, the graph F_0', and the two graphs obtained from the order-15 graph F_2 by adding a leaf at a hub or at the midpoint of its unique 2-edge branch, seven classes in all. For odd $n \geq 17$ the unique extremal graph is F_2. For even $n \geq 18$ there are exactly three: F_0', and the two graphs obtained from the order-$(n-1)$ graph F_2 by adding a leaf at a hub or at the midpoint of its unique 2-edge branch.

Proof. Write $\eta(H) = 3|V(H)| - 2e(H)$ for a triangle-free graph H. The argument is the cut-vertex analysis of Theorem 5.4, continued until the core is one graph from the table of Lemma 5.2.
Reduction to one core. Lemma 5.5 supplies a core Q of order h and the identity $3n - 2e(G) = \eta(Q) + n - h$.
Upper bound. Every fixed row of the table in Lemma 5.2 satisfies $\eta \geq h - 10$. Each such graph has order at most 10 except F_{12}, which has order 11 and $\eta = 3$, and $3 \geq 11 - 10$. The same inequality holds for K_1 and K_2, with deficits 3 and 4. For every such core, $\eta(Q) + n - h \geq n - 10$. The two parameterised rows are F_2, of odd order and deficit 5, and F_0', of even order and deficit 6. If the core is F_2 and the final order is even, then at least one vertex has been attached, so the displayed deficit is at least 6. If the core is F_2 and the final order is odd, the deficit is at least 5. If the core is F_0', the deficit is at least 6. In every case

$$3n - 2e(G) \geq \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

Attainment. The value $f(n)$ is attained by the graphs named in the statement; the next paragraph shows that no other graph attains it. A leaf added to a triangle-free strong C_4-free graph creates neither a triangle nor a cycle. Realise the Petersen graph on the 2-element subsets of $\{1, 2, 3, 4, 5\}$, with two subsets adjacent when they are disjoint. This is the graph P of Theorem 3.6. The symmetric group on the ground set acts on the vertices. The action is vertex-transitive and edge-transitive, and it carries any two nonadjacent vertices to $\{1, 2\}$ and $\{1, 3\}$. The path

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 3\}, \{1, 4\}$$

is an induced P_5 starting at $\{1, 2\}$, and

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 4\}, \{1, 3\}$$

is an induced P_5 with ends $\{1, 2\}$ and $\{1, 3\}$. Thus every vertex of P is an end of an induced P_5, and every pair of nonadjacent vertices is the pair of ends of an induced P_5.
Place a leaves at one end of an edge of P and b leaves at the other, with $a + b = n - 10$ and $a \geq b \geq 0$. The case $b = 0$ is the placement at a single vertex. The graph is connected, triangle-free and strong C_4-free. It has $15 + (n - 10)$ edges, so its deficit is $n - 10$. An induced path uses a leaf only as an end. A path that uses two leaves at the same vertex has order three. A path that uses one leaf at each end of the chosen edge is the edge itself with the two leaves attached: the ends of that edge are adjacent, so no longer induced path of P has those ends. A path that uses one leaf restricts to an induced path of P and has order at most six, because P is P_6-free. Thus the graph is P_7-free. Vertex-transitivity and edge-transitivity give one isomorphism class for each pair (a, b). Distinct pairs give nonisomorphic graphs: the Petersen graph is cubic, so the degrees of the vertices adjacent to a leaf are $3 + a$ and $3 + b$ when $b \geq 1$, and the single value $3 + a$ when $b = 0$. These are the classes in $\mathcal{P}(n)$, and they attain deficit $n - 10$.
For the deficit $5 + \mathbf{1}_{\{n \text{ even}\}}$, use F_2 at odd orders and F_0' at even orders. In the descriptions preceding Lemma 5.2, each of these graphs consists of internally disjoint three-edge paths between two hubs, together with one additional two-edge path in the case of F_2. Every cycle uses two of those paths and therefore has length at least five. An induced path that contains at most one hub meets that hub in at most two of the paths and so has at most five vertices. An induced path that contains both hubs uses at most one path between them, a segment of at most three edges, and each end extends by at most one further edge before the next vertex meets the opposite hub and creates a chord. Its order is at most six. The table of Lemma 5.2 gives deficits 5 and 6 respectively. The smaller of the two deficits is therefore attained.
Equality. Write $\delta(n)$ for the minimum in the statement, and let G satisfy $3n - 2e(G) = \delta(n)$. The identity $3n - 2e(G) = \eta(Q) + n - h$ and the upper bound force the core Q to be the Petersen graph, F_2 or F_0'. Every other core in the table of Lemma 5.2, and each of K_1 and K_2, has $\eta(Q) - h \geq -8$, hence deficit at least $n - 8$. For $n \geq 10$ one has $n - 8 > \delta(n)$, since $\delta(n) \leq n - 10$.

If Q is the Petersen graph, then $\eta(Q) = 0$, $h = 10$ and the deficit is $n - 10$. This equals $\delta(n)$ if and only if $10 \le n \le 16$. The vertices outside Q form trees attached at single vertices of Q. A tree of depth at least two contains a path y, a, s of length two ending at its attachment vertex s. Concatenating that path with an induced P_5 of Q that starts at s produces an induced P_7: the only edge from $\{y, a\}$ into Q is as, and a is not adjacent to the neighbour of s on the P_5 because that neighbour and a would form a triangle with s. Thus every attached tree is a star of leaves at its attachment vertex. If leaves are attached at two nonadjacent vertices u and v, an induced P_5 with those ends, extended by one leaf at each end, is an induced P_7. The set of attachment vertices is therefore a clique. The Petersen graph is triangle-free, so the attachments lie on one vertex or on the two ends of one edge. These are the classes in $\mathcal{P}(n)$.
If $Q = F_2$, then $\eta(Q) = 5$ and $h = 5 + 2t$ is odd, so the deficit is $5 + n - h$. For odd n this equals $\delta(n) = 5$ if and only if $n = h$ and $n \ge 15$, in which case G is F_2. For even n the orders n and h have opposite parity, so $n - h \ge 1$ and the deficit is at least 6. Equality with $\delta(n)$ holds if and only if $n - h = 1$ and $\delta(n) = 6$, hence $n \ge 16$ is even and G is F_2 of order $n - 1$ with a single leaf. Here $t = (n - 6)/2 \ge 5$.
If $Q = F_0'$, then $\eta(Q) = 6$ and the deficit is $6 + n - h \ge 6$, with equality if and only if $n = h$. Then $\delta(n) = 6$, so $n \ge 16$ is even and G is F_0'.
It remains to decide where the leaf may be placed on F_2. Adding a leaf at a vertex v of a P_7-free graph preserves P_7-freeness if and only if v is not an end of an induced P_6: the leaf has degree one, so it is an end of every induced path that contains it, and the remainder is an induced path starting at v. Label F_2 as in the description before Lemma 5.2, and call v_1 and v_3 the hubs. The unique 2-edge branch is $v_1v_2v_3$, with midpoint v_2. The remaining v_1–v_3 paths are the t matching paths through $A_1 \cup A_3$ and the path $v_1v_5v_4v_3$; call these the long branches. The map that swaps v_1 with v_3, swaps v_5 with v_4, fixes v_2, and swaps the two ends of each matching edge preserves the cycle, the matching, and every spoke, so it is an automorphism. It exchanges the two hubs, so a leaf at either hub gives one isomorphism class.
The hubs and v_2 are not ends of an induced P_6, and every other vertex is. Let P be an induced path starting at v_1. If the second vertex is v_2, the third is v_3, and the fourth is either v_4 or a vertex of A_3. In the first subcase the only continuation is v_5, and v_1v_5 is a chord, so the induced path has order at most four. In the second the only continuation is the match in A_1, and that match is adjacent to v_1, so again the order is at most four. If instead the path leaves v_1 along a long branch, it may continue to v_3 and then to the v_3-end of one further long branch. The resulting path has order five. Its last vertex has no neighbour off that path except its partner on the same long branch, and that partner is adjacent to v_1. The path does not extend, and the edge v_1v_2 is a chord of any continuation through v_2. Thus no induced P_6 starts at v_1, and none starts at v_3. Every induced path that starts at v_2 has second vertex v_1 or v_3. From v_1 it can enter only one long branch, and the far end of that branch is v_3, which is adjacent to v_2. The path stops before v_3 and has order at most four. Thus v_2 is not an end of an induced P_6.
Now let p be an internal vertex of a long branch, adjacent to v_1; the automorphism gives the vertices adjacent to v_3. Since $t \ge 5$, two further long branches v_1abv_3 and v_1cdv_3 are available. The path p, v_1, a, b, v_3, d is an induced P_6. The only neighbours of an internal vertex of a long branch are its two neighbours on that branch, the hubs are nonadjacent, and vertices on distinct long branches are nonadjacent except through the hubs. Thus p is an end of an induced P_6.
A leaf at a hub or at v_2 therefore keeps the graph P_7-free, and a leaf at any other vertex does not. The two hubs are equivalent under the automorphism above, while v_2 is the unique common neighbour of the hubs, so the two placements are not isomorphic: for $t \ge 5$ the neighbour of the leaf has degree $t + 3$ or degree 3 according as it is a hub or v_2.
The graphs just listed are pairwise nonisomorphic. A graph in $\mathcal{P}(n)$ has $n - 10$ vertices of degree one and its 2-connected core has order 10. The graph F_2 has minimum degree two. The graph F_0' has minimum degree two and even order. Each one-leaf extension of F_2 has exactly one vertex of degree one, and the two extensions are separated by the degree of its neighbour. For $10 \le n \le 14$ one has $\delta(n) = n - 10$, so only $\mathcal{P}(n)$ occurs. For $n = 15$, $\mathcal{P}(15)$ consists of the three pairs $(5, 0)$, $(4, 1)$ and $(3, 2)$, and F_2 is a fourth class. For $n = 16$, $\mathcal{P}(16)$ consists of the four pairs $(6, 0)$, $(5, 1)$, $(4, 2)$ and $(3, 3)$, and F_0' together with the two one-leaf extensions makes seven classes. For odd $n \ge 17$ only F_2 meets $\delta(n)$. For even $n \ge 18$ the three graphs F_0' and the two one-leaf extensions are the only ones. □

An enumeration of the equality graphs of orders 10 through 18 finds $1, 1, 2, 2, 3, 4, 7, 1, 3$ isomorphism classes. The count agrees with Theorem 5.6 and is not used in the proof.

Theorem 5.7. *Let G be a connected strong C_4-free P_7-free graph with at most one triangle. Then $S(G) \le 3n$. In particular the same bound holds for a graph with exactly one triangle.*

Proof. Suppose G is a counterexample of minimum order in the stated class, so $S(G) \ge 3n + 1$. Theorem 5.4 forces G to contain a triangle, hence exactly one. The short-side argument at a cut vertex in Theorem SB.1 applies inside this hereditary class. Let v be a cut vertex. For each component C of $G - v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ with end v. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \le 6$ vertices, so some component C_1 has $\rho_{C_1} \le 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \ge 2$.

Both graphs are connected, smaller than G, strong C_4-free and P_7-free, and each has at most one triangle. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3,

$$S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus G is 2-connected.

Let v be a vertex of minimum degree among the three vertices of the triangle, write $d = d(v)$, and let u and w be the other two. The graph $H = G - v$ is connected and triangle-free. Apply Lemma 5.5 to H, and write Q for its core, $h = |V(Q)|$ and $\eta = \eta(Q)$. Set

$$a = \big|N_G(v) \cap V(Q)\big|, \qquad c = \big|V(H) \setminus \big(V(Q) \cup N_G(v)\big)\big|,$$

and $t = |V(H)| - h$. The vertices of H outside Q are the $d - a$ neighbours of v that miss Q, together with the c vertices counted above, so $t = d - a + c$. Lemma 5.5 gives $e(H) = e(Q) + t$. Since G has one triangle, $S(G) = 2e(G) - 3$ and $e(G) = e(H) + d$, and therefore

$$S(G) - 3n = 2d - 6 - \eta - t = d + a - 6 - \eta - c. \tag{5.1}$$

The core satisfies $a \le \eta + 3$ and, for every edge xy of Q,

$$a + \min\{d_Q(x), d_Q(y)\} \le \eta + 5. \tag{5.2}$$

These bounds are read from the list in Lemma 5.2, using the descriptions that precede that lemma. If Q is the Petersen graph, then $\eta = 0$. Every two nonadjacent vertices of the Petersen graph have a common neighbour, so two such vertices cannot both be adjacent to v. Thus $N(v) \cap V(Q)$ is a clique, necessarily of size at most two, and every degree in Q is three. Both bounds follow. For each remaining fixed graph of order at least three one has $h \le 11$ and minimum degree at least two. The open neighbourhoods in Q of any two vertices of $N(v) \cap V(Q)$ are disjoint: a common neighbour would form a 4-cycle with v. Hence $a \le \lfloor h/2 \rfloor \le 5$. Every degree in these graphs is at most four. The only rows with $\eta < 4$ are F_4 and F_{12}, both with $\eta = 3$. The path-and-spoke description of F_4 has maximum degree three. In F_{12} the only vertices of degree four are v_1 and v_5 on the 7-cycle. Each is adjacent to two cycle vertices and to two of $\{x_1, x_2, x_5\}$, and v_1 is not adjacent to v_5, so every edge of F_{12} has an end of degree at most three. This gives (5.2) and the bound on a. In each parameterised family every non-hub vertex is adjacent to a hub, and each hub has at most one neighbour in common with v, so $a \le 4$ even after both hubs are counted. Every edge has an end of degree two, because the edges are the internal edges of the paths between the hubs described above, and η is 5 or 6. The graphs K_1 and K_2 satisfy both bounds directly: K_1 has $\eta = 3$ and no edge, and K_2 has $\eta = 4$.

If u or w lies outside Q, say u, the tree attachments give u at most one neighbour in Q. Apart from w, no neighbour of u in H is adjacent to v, because the triangle is unique. Thus $d \le d(u) \le c + 3$. Substituting in (5.1) gives $S(G) - 3n \le a - \eta - 3$, and $a \le \eta + 3$ yields $S(G) - 3n \le 0$. If instead both u and w lie in Q, then uw is an edge of Q, and every neighbour of either vertex that lies in $H - V(Q)$ lies in the set counted by c. Since v has minimum degree among the three triangle vertices,

$$d \le 1 + c + \min\{d_Q(u), d_Q(w)\}.$$

Substituting in (5.1) and applying (5.2) to the edge uw again yields $S(G) \le 3n$. □

Lemma 5.8. *Let G be a connected strong C_4-free P_7-free graph that contains a triangle, satisfies $L(v) \ge 4$ for every vertex, and contains no two adjacent vertices of degree two. Then G is isomorphic to the graph $H_{3,2}$ obtained by partitioning the vertices of K_5 into parts of sizes three and two and subdividing each of the six edges with one end in each part exactly once. This graph has*

$$(n, e, T, S) = (11, 16, 1, 29).$$

Proof. Let A be the part of size three, let B be the part of size two, and let Z be the set of six subdivision vertices. The construction puts a triangle on A, an edge on B, no edge between A and B, and exactly one vertex of Z on each pair with one end in A and one end in B. Thus $n = 11$ and $e = 3 + 1 + 12 = 16$. A vertex of Z is adjacent to one vertex of A and one vertex of B, which are nonadjacent, and no vertex of Z lies in a triangle with an edge of A or with the edge of B. The only triangle is the triangle on A, so $T = 1$ and $S = 2e - 3T = 29$. Each vertex of A has degree four and lies in one triangle, so its deletion loss is 5. Each vertex of B has degree four and lies in no triangle, so its deletion loss is 8. Each vertex of Z has degree two, with nonadjacent neighbours, so its deletion loss is 4. No two vertices of Z are adjacent. Figure 3 draws the graph.

There is no 4-cycle. The subgraph induced by $A \cup B$ is a triangle together with a disjoint edge. A cycle through a vertex $s \in Z$ uses both edges at s, so it contains the path from its neighbour in A to its neighbour in B. Those two vertices have no common neighbour other than s.

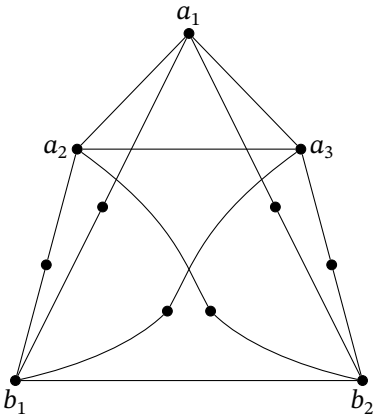


FIGURE 3 | The graph $H_{3,2}$ of Lemma 5.8. The part $\{a_1, a_2, a_3\}$ spans the unique triangle, the part $\{b_1, b_2\}$ spans the remaining edge of K_5, and each of the six edges of K_5 between the parts is subdivided once. The two curved paths are the subdivisions of a_2b_2 and a_3b_1; they do not meet.

There is no induced P_7. The set A is a clique and the set B is a clique, so an induced path contains at most two vertices of A and at most two vertices of B. An induced path on seven vertices would therefore contain at least three vertices of Z. A vertex of Z on an induced path is an end of the path, or else lies between its two neighbours. Suppose two vertices of Z are internal. If their neighbours in B are distinct, those two vertices of B are adjacent, hence consecutive on the path, and the path contains a subpath $a\,s\,b\,b'\,s'\,a'$ with $a, a' \in A$. If their neighbours in B coincide, the path contains a subpath $a\,s\,b\,s'\,a'$ with $a, a' \in A$. In both cases the edge aa' is a chord. Thus at most one vertex of Z is internal. At least three vertices of Z lie on the path, and a path has only two ends, so the path contains exactly three vertices of Z: the two ends and one internal vertex. The other four vertices lie in $A \cup B$.

The internal vertex contributes a subpath asb with $b \in B$. Suppose both vertices of B lie on the path. They are consecutive, so the path contains the block $asbb'$. The fourth vertex of $A \cup B$ on the path is a vertex $a' \in A$, and $asbb'$ is a block of four consecutive vertices. There is no edge between A and B, so a' is not adjacent to b'. The only remaining possibility is that a' is adjacent to a on the path. The two ends of the resulting seven-vertex path lie in Z. The end adjacent to a' has its other neighbour in B, because its neighbour on the path lies in A. Both vertices of B are already on the path and neither is the successor of that end, so the other neighbour is a chord. Suppose instead that only one vertex of B lies on the path. Then all three vertices of A lie on the path. Those three vertices form a clique, so they do not lie on an induced path. This contradiction shows that $H_{3,2}$ is strong C_4-free and P_7-free, and the local hypotheses have already been checked. Thus $H_{3,2}$ is a graph of the kind named in the lemma.

It remains to show that no other isomorphism type occurs. Let G satisfy the hypotheses and fix an induced triangle as the starting subgraph, still called H during the growth. A labelled induced subgraph is enlarged by the following rule. If some vertex v of the current graph H has deletion loss $L_H(v) < 4$, computed in H, and v has no neighbour in G outside H, then the degree and the triangle count of v have attained their values in G, so $L_G(v) < 4$. If a and b are adjacent vertices of degree two in H, then at least one of them has a neighbour in G outside H. In either situation the rule selects a nonempty set $X \subseteq V(H)$ that some new vertex must meet: the deficient vertex, or one vertex of the degree-two pair. If H already satisfies $L_H \geq 4$ at every vertex and has no two adjacent vertices of degree two, its isomorphism type is compared with $H_{3,2}$. If that graph is still a proper induced subgraph of G, connectivity supplies a vertex outside H with a neighbour in H, and every nonempty neighbourhood is admitted at that node.

For the next labelled vertex z, form every set $U \subseteq V(H)$ that meets the required set X, or every nonempty U at an unrestricted node, and join z precisely to U. Keep the child when the new graph has neither a 4-cycle subgraph nor an induced P_7. A new 4-cycle exists precisely when two vertices of U have a common neighbour in the current graph. An induced P_7 created by the addition uses z and six vertices of the current graph. Fix any embedding of the current graph into G. The neighbourhood, in the current vertex set, of a vertex that the rule requires is one of these sets U, and the child it determines is kept. An embedded proper prefix therefore extends. Every graph satisfying the hypotheses arises along some branch, with no restriction on its order.

The search tree is finite. Section A of the supplement records the audit, and the program `verify_gen3.py` checks the same certificate. The tree has 169 labelled nodes and 52 leaves. By order $3, \ldots, 17$ the numbers of visited nodes are

$$1, 1, 1, 1, 2, 3, 5, 14, 23, 24, 23, 19, 24, 20, 8.$$

Four nodes are unrestricted. All four have order 11, and each is the graph $H_{3,2}$. The largest order reached is 17. That bound is an output of the search: the verifier imposes no order cutoff and performs no isomorphism pruning. Every branch other than those four nodes is closed by a local obstruction, and every one-vertex extension of $H_{3,2}$ leaves the class or violates a local hypothesis. Therefore $G \cong H_{3,2}$. □

Theorem 5.9. *Every connected strong C_4-free P_7-free graph G satisfies $S(G) \leq 3n$, with equality if and only if G is the Petersen graph.*

Proof. The triangle-free case, including the equality, is Theorem 5.4. Suppose G contains a triangle and $S(G) \geq 3n$, and choose such a connected graph of minimum order. Every smaller connected graph in the class satisfies $S \leq 3|V|$, and the inequality is strict whenever the smaller graph contains a triangle.
Suppose s is a cut vertex. Induced paths that start at s in distinct components of $G - s$ concatenate. Since $p(G) \leq 6$, some component determines a side G_1, induced by that component together with s, in which every induced path starting at s has order at most three. Let G_2 be the subgraph induced by the remaining vertices together with s, and write $n_i = |V(G_i)|$. Then $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$. The graph G_2 is a smaller connected graph in the class. If it contains a triangle, minimality gives $S(G_2) < 3n_2$; if it does not, Theorem 5.4 gives $S(G_2) \leq 3n_2$. Lemma 2.3 therefore yields

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus G has no cut vertex, and $G - v$ is connected for every vertex v.
Suppose $L(v) \leq 3$ for some vertex v. Then $S(G - v) = S(G) - L(v) \geq 3n - 3 = 3(n - 1)$. The graph $G - v$ is connected and lies in the class. It contains no triangle: otherwise minimality would give the strict inequality $S(G - v) < 3(n - 1)$. Theorem 5.4 therefore forces $G - v$ to be the Petersen graph, so $n = 11$ and $S(G - v) = 30$. Any two nonadjacent vertices of the Petersen graph have a common neighbour, so the neighbours of v form a clique. The Petersen graph is triangle-free, and a clique in it has size at most two. The graph G contains a triangle, so v is adjacent to both ends of an edge and to no further vertex of $G - v$. Then $d(v) = 2$, $t(v) = 1$ and $L(v) = 2 \cdot 2 - 3 \cdot 1 = 1$, hence $S(G) = 31 < 33$. This contradiction shows that $L(v) \geq 4$ for every vertex v.
Suppose u and w are adjacent vertices of degree two. The bound on the deletion loss forces $t(u) = t(w) = 0$, so neither vertex lies in a triangle and the two neighbours of u are nonadjacent. Let x be the neighbour of u other than w, and let y be the neighbour of w other than u. Then $x \neq y$, the edge xy is absent, and x is not adjacent to w. Deleting $\{u, w\}$ removes the three edges xu, uw and wy and preserves every triangle. The graph that remains may be disconnected. Its number of edges is $e(G) - 3$ and its number of triangles is $T(G)$, so its value of S is $S(G) - 6 \geq 3n - 6 = 3(n - 2)$. For a disconnected graph, S is the sum of the values on the components. Each component is a smaller connected graph in the class. The triangle of G survives in one component, and minimality makes the inequality strict for that component, while every other component contributes at most three times its order. The sum is therefore strictly less than $3(n - 2)$, a contradiction.
Thus G satisfies the hypotheses of Lemma 5.8. That lemma gives $n = 11$ and $S(G) = 29 < 33$, which is impossible. No connected graph in the class with a triangle satisfies $S \geq 3n$. Combined with Theorem 5.4, this is the stated bound, and equality holds only for the Petersen graph. □

Lemma 5.10. *Every connected strong C_4-free P_7-free graph with at least two triangles has a vertex v with $L(v) \leq 3$.*

Proof. Let G be connected, strong C_4-free and P_7-free, and suppose G has at least two triangles. Two triangles do not share an edge. If they did, their four or three vertices would contain a 4-cycle: the two vertices opposite a shared edge, together with the ends of that edge, span a 4-cycle when the opposite vertices are nonadjacent, and a shared edge whose opposite vertices are adjacent still leaves a 4-cycle through one opposite vertex and the third vertex of either triangle. Thus two triangles share at most one vertex.
If two triangles share a vertex, an extra edge between their unshared vertices creates a 4-cycle with an edge of one triangle. The union of the two triangles is therefore an induced subgraph, on five vertices. If two triangles are disjoint, let d be the length of a shortest path joining their vertex sets. A second edge between the two triangles creates a 4-cycle, whether or not the two edges share an end. For $d \geq 2$, no interior vertex of the path is adjacent to a second vertex of either triangle: such an adjacency would be a chord of the shortest path, or else it would meet two vertices of one triangle and form a 4-cycle with the third. The two triangles and the path therefore induce the graph consisting of the two triangles joined by that path. Extending the path by one further vertex of each triangle produces an induced path on $d + 3$ vertices. Since $p(G) \leq 6$, one has $d \leq 3$.
Four induced subgraphs therefore exhaust the configurations. Two triangles may share a vertex, or they may be joined by a path of length 1, 2 or 3. These graphs have orders 5, 6, 7 and 8 respectively. From each of them, grow a labelled induced subgraph by the rule of Lemma 5.8, with one change: at every node select a single vertex whose current deletion loss is strictly less than 4, and admit only those neighbourhoods that meet the corresponding one-vertex set. Section A records

the four searches. They visit 36, 31, 46 and 5 nodes, and the largest orders they reach are 17, 15, 11 and 10. Every branch closes. An induced copy of any of the four graphs inside a graph with $L \geq 4$ at every vertex would extend at every node. Therefore no graph with at least two triangles has $L \geq 4$ at every vertex. □

Remark 5.11. Lemma 5.10 is not required for Theorem 5.9. It yields a second proof of the inequality $S \leq 3n$. Proceed by induction on the order. Theorem 5.4 and Theorem 5.7 give the inequality when the graph has at most one triangle. If there are at least two triangles, Lemma 5.10 supplies a vertex v with $L(v) \leq 3$. Each component of $G - v$ is smaller and lies in the class, so the inductive bound sums to $S(G - v) \leq 3(n - 1)$. Hence $S(G) \leq 3(n - 1) + 3 = 3n$. The equality analysis remains that of Theorem 5.9.

Corollary 5.12. *Let G be connected and strong C_4-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then G contains a triangle.*

Proof. If G were triangle-free, Theorem 5.4 would give $S(G) \leq 3n$. □

Lemma 5.13. *Let G be a graph and let K be a clique cutset of order t, with $G - K$ having $k \geq 2$ components. For each component C let G_C be the subgraph induced by $C \cup K$, and write $\Delta(H) = 2|V(H)| - 3 - e(H)$. Then*

$$\Delta(G) = \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].$$

The bracket equals 1 if $t = 1$ and equals 0 if $t = 2$ or $t = 3$.

Proof. The edges inside K are counted k times among the graphs G_C, and $|V(G)| = \sum_C |V(G_C)| - (k-1)t$, so

$$e(G) = \sum_C e(G_C) - (k-1)\binom{t}{2}.$$

Therefore

$$\begin{aligned}
\Delta(G) &= 2\sum_C |V(G_C)| - 2(k-1)t - 3 - \sum_C e(G_C) + (k-1)\binom{t}{2} \\
&= \sum_C \big(\Delta(G_C) + 3\big) - 3 - 2(k-1)t + (k-1)\binom{t}{2} \\
&= \sum_C \Delta(G_C) + (k-1)\Big[\binom{t}{2} + 3 - 2t\Big].
\end{aligned}$$

The bracket is $0 + 3 - 2 = 1$ for $t = 1$, $1 + 3 - 4 = 0$ for $t = 2$, and $3 + 3 - 6 = 0$ for $t = 3$. □

Lemma 5.14. *Let G be a connected strong C_4-free P_7-free graph on $n \geq 2$ vertices, and write $\Delta(G) = 2n - 3 - e(G)$. Then $\Delta(G) \geq 0$. If $n \geq 4$, then $\Delta(G) \geq 1$, so $e(G) \leq 2n - 4$ and $S(G) \leq 4n - 8 - 3T(G)$. In particular $\ell(G) < 4$.*

Proof. A copy of K_4 contains a 4-cycle, so $\omega(G) \leq 3$. A diamond contains a 4-cycle, so every graph in the class is $(P_7, C_4, \text{diamond})$-free in the sense of Chen, Wu and Xu [4]. The proof is by induction on n. For $n = 2$ the graph is K_2 and $\Delta = 0$. For $n = 3$ the maximum of e is 3, attained by K_3, and $\Delta \geq 0$.

Let $n = 4$. A graph on four vertices with five edges is K_4 minus an edge. Labelling the missing edge cd, the cycle $a\,c\,b\,d$ uses four edges of the graph. Thus $e \leq 4$ and $\Delta = 5 - e \geq 1$.

Now let $n \geq 5$, and assume the two claims for every smaller graph in the class. Every block G_C arising from a clique cutset is itself connected, strong C_4-free and P_7-free, of order strictly less than n.

Suppose first that v is a cut vertex, and that $G - v$ has $k \geq 2$ components. Lemma 5.13 with $t = 1$ gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. Each side has order at least 2, so $\Delta(G_i) \geq 0$ by induction, and $\Delta(G) \geq k - 1 \geq 1$.

Suppose next that a triangle K is a clique cutset and $G - K$ has $k \geq 2$ components. Each side G_i contains K and at least one further vertex, so its order is at least 4. Lemma 5.13 gives $\Delta(G) = \sum_i \Delta(G_i)$, and induction supplies $\Delta(G_i) \geq 1$. Hence $\Delta(G) \geq k \geq 2$.

Suppose that $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. Lemma 5.13 gives $\Delta(G) = \sum_i \Delta(G_i)$. Induction gives $\Delta(G_i) \geq 0$ for every side, and each side has order at least 3. If every side had deficit 0, then each side would have order 3 and exactly three edges, so each side would be a triangle. Two triangles that share the edge uv span a 4-cycle on their two opposite vertices, which contradicts the hypothesis that G is strong C_4-free. Some side therefore has positive deficit, and $\Delta(G) \geq 1$.

Suppose finally that G has no clique cutset. Theorem 1.1 of [4] states that a connected $(P_7, C_4, \text{diamond})$-free graph with no clique cutset is the Petersen graph, or the graph F defined below, or else has minimum degree at most $\max\{2, \omega - 1\}$. Here $\omega \leq 3$, so the last quantity is at most 2. The Petersen graph has $n = 10$ and $e = 15$, hence $\Delta = 2$. The graph F is

obtained from a 7-cycle $v_1 \cdots v_7$ by adding an independent set $\{y_1, y_2, y_3\}$ with

$$N(y_1) = \{v_1, v_4, v_5\}, \qquad N(y_2) = \{v_2, v_5, v_6\}, \qquad N(y_3) = \{v_3, v_6, v_7\}.$$

The three triangles of F are $y_1v_4v_5$, $y_2v_5v_6$ and $y_3v_6v_7$, and F has $n = 10$ and $e = 16$. The vertices $v_1, v_7, y_3, v_3, v_4, v_5, y_2$, in that order, induce a P_7. The path edges are the cycle edges v_1v_7, v_3v_4 and v_4v_5, together with v_7y_3, y_3v_3 and v_5y_2. There is no chord. The set $\{y_1, y_2, y_3\}$ is independent; y_3 meets the cycle only in $\{v_3, v_6, v_7\}$ and y_2 only in $\{v_2, v_5, v_6\}$; and none of v_1v_3, v_1v_4, v_1v_5, v_7v_3, v_7v_4, v_7v_5, v_3v_5 is a cycle edge. Thus F is not P_7-free. It is not a graph of the class, and it contributes no candidate. Every remaining graph in the class with no clique cutset is the Petersen graph or has a vertex of degree at most 2. The bound on the number of edges is unchanged. If some vertex v has degree at most 2, then v is not a cut vertex, so $G - v$ is connected and lies in the class. Deleting v changes the deficit by

$$\Delta(G) - \Delta(G - v) = 2 - d(v) \geq 0.$$

For $n \geq 5$ the order of $G - v$ is at least 4, so induction gives $\Delta(G - v) \geq 1$ and therefore $\Delta(G) \geq 1$.
Lemma 2.1(3) gives $S = 2e - 3T$. For $n \geq 4$ one has $e \leq 2n - 4$, so $S \leq 4n - 8 - 3T \leq 4n - 8$ and $\ell \leq 4 - 8/n < 4$. For $2 \leq n \leq 3$ one has $e \leq 2n - 3$ and $S \leq 4n - 6$, so again $\ell < 4$. □

Lemma 5.15. *Let G be a connected strong C_4-free P_7-free graph.*

1. *If $n \geq 5$ and $T(G) \leq 1$, then $\Delta(G) \geq 2$, so $e(G) \leq 2n - 5$. If $T(G) = 1$, then $S(G) \leq 4n - 13$.*
2. *If $n \geq 4$ and $T(G) \geq 2$, then $S(G) \leq 4n - 14$.*
3. *If $n \geq 5$ and $T(G) \geq 1$, then $S(G) \leq 4n - 13$.*
4. *If $n \geq 6$ and $T(G) \leq 2$, then $\Delta(G) \geq 2$.*

Proof. If $T \geq 2$ and $n \geq 4$, Lemma 5.14 gives $e \leq 2n - 4$, hence $S = 2e - 3T \leq 2(2n - 4) - 6 = 4n - 14$. This is (2). Once (1) is known, $T = 1$ gives $S = 2e - 3 \leq 4n - 13$, and (2) is at least as strong, so (3) follows. It remains to prove $\Delta \geq 2$ when $n \geq 5$ and $T \leq 1$.
The proof is by induction on n. Let $n = 5$. A chordless cycle in a strong C_4-free graph has length at least 5, so the only possible hole on five vertices is a 5-cycle. A chord of C_5 produces a 4-cycle, and there is no room for a further vertex, so $G \cong C_5$. Then $e = 5$, $T = 0$ and $\Delta = 2$. If G has no hole, then G is chordal. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and its neighbourhood is a clique. That clique has order at most two: a simplicial vertex of degree three or more, together with three neighbours, contains a K_4 and therefore a 4-cycle. Deleting a simplicial vertex of degree one removes one edge and no triangle. Deleting a simplicial vertex of degree two removes two edges and exactly one triangle. Repeating until a single vertex remains gives $e = n - 1 + T$. Here $T \leq 1$, so $e \leq 5$ and $\Delta = 7 - e \geq 2$.
Now let $n \geq 6$, with $T(G) \leq 1$, and assume the deficit claim for every smaller graph in the class that has at most one triangle. By Lemma 5.14 every graph in the class of order at least 2 has nonnegative deficit, and every graph of order at least 4 has deficit at least 1.
Suppose v is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma 5.13 gives $\Delta(G) = \sum_i \Delta(G_i) + (k - 1)$. If this quantity were at most 1, then $k = 2$ and both sides would have deficit 0. A side of deficit 0 has order at most 3, so $n \leq 3 + 3 - 1 = 5$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose a triangle is a clique cutset, with $k \geq 2$ sides. Each side has order at least 4, so each has deficit at least 1 by Lemma 5.14, and Lemma 5.13 gives $\Delta(G) \geq 2$.
Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components, and suppose toward a contradiction that $\Delta(G) \leq 1$. Lemma 5.13 gives $\Delta(G) = \sum_i \Delta(G_i)$. A side of order at least 4 has deficit at least 1. A side of deficit 0 therefore has order 3 and three edges, so it is a triangle. Two such sides would give two triangles and a 4-cycle, which is impossible under $T(G) \leq 1$ and the strong C_4-free hypothesis. If every side had deficit at least 1, the sum would be at least $k \geq 2$. Hence exactly one side is a triangle, of deficit 0, and exactly one other side is present: a third side would contribute deficit at least 1 and would push the sum to at least 2. The hypothesis $T(G) \leq 1$ forces the other side G_2 to be triangle-free. Its order is $n - 1 \geq 5$. The inductive hypothesis gives $\Delta(G_2) \geq 2$, so $\Delta(G) \geq 2$, a contradiction. Thus $\Delta(G) \geq 2$.
Suppose finally that G has no clique cutset. The Petersen graph has $T = 0$ and $\Delta = 2$. The graph F displayed in the proof of Lemma 5.14 is not P_7-free, so it is not a candidate in this class. In every remaining case Chen, Wu and Xu supply a vertex v of degree at most 2. This vertex is not a cut vertex. It lies in at most one triangle, and $T(G - v) = T(G) - t(v) \leq 1$. The graph $G - v$ is connected, strong C_4-free and P_7-free, of order at least 5. Induction gives $\Delta(G - v) \geq 2$. Since $\Delta(G) = \Delta(G - v) + 2 - d(v) \geq \Delta(G - v)$, one has $\Delta(G) \geq 2$.
It remains to prove (4). Part (1) already gives the claim when $T \leq 1$. Suppose $T = 2$, and proceed by induction on n from order six. Lemma 5.14 supplies $\Delta \geq 0$ for every connected graph in the class of order at least two, and $\Delta \geq 1$ for every such graph of order at least four.

Suppose v is a cut vertex and $G - v$ has $k \geq 2$ components. Lemma 5.13 gives $\Delta(G) = \sum_i \Delta(G_i) + (k-1)$. A value at most 1 would require $k = 2$ and both sides of deficit 0, hence both of order at most three, and then $n \leq 5$. Thus $\Delta(G) \geq 2$. Suppose a triangle is a clique cutset. Each side has order at least four, the side deficits add, and Lemma 5.14 gives $\Delta(G) \geq 2$. Suppose $\{u, v\}$ is a clique cutset and $G - \{u, v\}$ has $k \geq 2$ components. The side deficits again add. A side of deficit 0 has order three and three edges, so it is a triangle. Two such sides would contain a 4-cycle, so there is at most one. If $\Delta(G) \leq 1$, there are exactly two sides, one a triangle and the other a graph of order $n - 1 \geq 5$ with at most one triangle. Part (1) gives that side deficit at least two, a contradiction. This handles every clique cutset.
Suppose G has no clique cutset. The Petersen graph does not have exactly two triangles. The graph F of Lemma 5.14 is excluded by the induced P_7 recorded there, so it is not a candidate with $T = 2$. Theorem 1.1 of Chen, Wu and Xu, in the form already used for Lemma 5.14, therefore supplies a vertex x of degree at most two. The graph $J = G - x$ is connected, and $\Delta(G) = \Delta(J) + 2 - d(x)$. For $n \geq 7$ the inductive hypothesis gives $\Delta(J) \geq 2$ when $T(J) \leq 2$, while $T(J) \leq 1$ is the case of part (1); in both subcases $\Delta(G) \geq 2$. For $n = 6$ the same comparison leaves only the case $d(x) = 2$, $|V(J)| = 5$ and $\Delta(J) = 1$. Then J has six edges. A hole in J would be a 5-cycle, and any chord of that cycle is a 4-cycle, so J is chordal. The simplicial deletion in the order-five argument above gives $e(J) = |V(J)| - 1 + T(J)$, hence $T(J) = 2$. Two triangles on five vertices cannot share an edge, for that graph contains a 4-cycle, and they cannot be disjoint, so they share exactly one vertex. Every pair of vertices of this graph has a common neighbour. Joining x to any two of them creates a 4-cycle, which contradicts $d(x) = 2$. □

Proposition 5.16. *Let G be a connected strong C_4-free P_7-free graph with $S(G) > 3n$. Then $T(G) \geq 2$, $n \geq 17$ and $3 < \ell(G) \leq 4 - 16/n$. The same edge-deficit estimates place every counterexample to REAL-133 with $\ell \leq 4$ in the cell*

$$r = 3, \qquad p = 6, \qquad T \geq 2, \qquad n \geq 17, \qquad 3 < \ell \leq 4 - \frac{16}{n}.$$

This cell is an intermediate estimate of that reduction. It is empty.

Proof. The second assertion is the case division that preceded Theorem 5.9. Write $r = r(G)$ and $p = p(G)$, and suppose G is a connected strong C_4-free counterexample to REAL-133 with $\ell \leq 4$. Theorem 5.18 gives the inequality whenever $\ell \leq 3$ and whenever $r \leq 2$ and $\ell \leq 4$, so $\ell > 3$ and $r \geq 3$. A violation means $p \leq r + \lceil \ell \rceil - 1$. Since $\ell \leq 4$ one has $\lceil \ell \rceil \leq 4$, so $p \leq r + 3$. Theorem 2.4 gives $p \geq 2r - 1$, hence $r \leq 4$. If $r = 4$, then $p = 7$, and Corollary 4.4 gives $\ell \leq 2$. Thus $r = 3$ and $p \leq 6$. Theorem 2.4 gives $p \geq 5$. If $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$. Hence $p = 6$ and $S > 3n$.
Now let G be any connected strong C_4-free P_7-free graph with $S > 3n$, whether or not it is a counterexample to the conjecture. Theorem 5.4 and Theorem 5.7 force $T \geq 2$. If $n \leq 5$, part (2) of Lemma 5.15 and the bound $S \leq 4n - 6$ of Lemma 5.14 give $S \leq 3n$. If $T = 2$ and $n \geq 6$, part (4) of Lemma 5.15 gives $\Delta \geq 2$, so $e \leq 2n - 5$ and $S = 2e - 6 \leq 4n - 16$. If $T \geq 3$ and $n \geq 4$, Lemma 5.14 gives $S \leq 4n - 8 - 3T \leq 4n - 17$. In the remaining range $S \leq 4n - 16$, so $\ell \leq 4 - 16/n$. Combined with $S > 3n$ this forces $n \geq 17$. For every such order, $4 - 16/n < 4$, so $\ell \neq 4$. A counterexample with $\ell \leq 4$ has $p = 6$ and $S > 3n$, so it lies in the stated cell. Theorem 5.9 forbids $S > 3n$, and the cell contains no graph. □

Remark 5.17. The degree bound in Theorem 1.1 of Chen, Wu and Xu does not by itself yield $S \leq 3n$. Lemmas 5.14 and 5.15 are the deficit estimates that follow from that theorem once clique cutsets of order 1, 2 and 3 are restored. The graph F of Lemma 5.14 is excluded by an induced P_7, so only the Petersen graph is counted as an exception, and the bound on the number of edges is unchanged. The resulting estimate is $\ell < 4$. On the range $S > 3n$ it is the intermediate bound of Proposition 5.16. The proof of $S \leq 3n$ for every number of triangles is Theorem 5.9.

Theorem 5.18 (Theorem A)**.** *Let G be connected and strong C_4-free with $n \geq 2$. Then $p(G) \geq r(G) + \ell(G)$ whenever $\ell(G) \leq 3$, whenever $r(G) \leq 2$ and $\ell(G) \leq 4$, whenever $r(G) = 4$ and $\ell(G) \leq 4$, and whenever $r(G) \geq 5$ and $\ell(G) \leq 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \leq 2$.*

Proof. Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Corollary 4.4. For the inequality we need $p \geq r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \leq 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \leq 4$, then Lemma 3.3 gives $\ell \leq 2$, a contradiction. So $p \geq 5$.
- $r = 2$, $c = 4$: if $p \leq 5$, then Theorem 3.6 gives $\ell \leq 3$, a contradiction. So $p \geq 6$.
- $r = 3$, $c \leq 2$: Theorem 2.4 gives $p \geq 5$.
- $r = 3$, $c = 3$: if $p \leq 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \geq 6$. Theorem 3.6 gives $\ell \leq 3$ for every P_6-free graph in the class, which does not rule out $c = 3$.

- $r \geq 4$, $c \leq 3$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 3$.
- $r = 4$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 7$. If $p = 7$, Corollary 4.4 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$, Remark 4.6 gives the same inequality.
- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + c$.

□

Equality in REAL-133 occurs for C_5 ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and C_6 ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

Corollary 5.19. *Every connected graph G on at least two vertices with $\ell(G) \leq 4$ satisfies Conjecture 1.1. The conjecture remains open for graphs with $\ell > 4$.*

Proof. If G contains a 4-cycle, then $c_{C_4}(G) = 0$ and the conjecture reads $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced.
Suppose G is strong C_4-free, $\ell(G) \leq 4$, and $p(G) < r(G) + \ell(G)$. The case division in the proof of Proposition 5.16 gives $r(G) = 3$, $p(G) = 6$ and $S(G) > 3n$. Theorem 5.9 gives $S(G) \leq 3n$. This contradiction shows that no such graph exists.
The comparison does not reach $\ell > 4$. In that range Corollary 6.1 still leaves the statements GEN_K for $K \geq 4$ and $\mathrm{RAD}(r)$ for $r \geq 5$. □

Remark 5.20. The floor form $p \geq r + \lfloor \ell \rfloor$ of the conjecture also holds for every connected graph with $\ell < 5$. If G contains a 4-cycle, then $\ell^{\,c_{C_4}(G)} = 1$, so both the real form and the floor form reduce to $p \geq r + 1$, which follows from a geodesic. Suppose G is strong C_4-free and $4 \leq \ell < 5$, and suppose the floor form fails. Then $p \leq r + 3$. Theorem 2.4 gives $p \geq 2r - 1$, so $r \leq 4$. If $r \leq 3$, then $p \leq 6$, and Theorem 5.9 gives $\ell \leq 3$, contradicting $\ell \geq 4$. If $r = 4$, then $p = 7$, and Corollary 4.4 gives $\ell \leq 2$, again a contradiction. The real-valued conjecture for $\ell > 4$ remains open.

6 | The reduction above density four

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

GEN_K: every connected strong C_4-free graph G with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.

$\mathrm{RAD}(r)$: every connected strong C_4-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Corollary 4.4 is the case $r = 4$ of $\mathrm{RAD}(r)$, in the stronger form $\ell \leq 2$.

Corollary 6.1. *Let G be a connected strong C_4-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \geq 3$. Then either*
(a) $r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so G violates GEN_K; or
(b) $r(G) = K + 1 \geq 4$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so G violates $\mathrm{RAD}(K + 1)$.

Proof. Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 3$ by Theorem 5.18. A violation means $p \leq r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \leq p \leq r + K$, so $r \leq K + 1$. If $r \leq K$, then $p \leq r + K \leq 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \leq p \leq 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. □

Since GEN_K ignores the radius, it is stronger than alternative (a). The statement GEN_K might fail for large K even if REAL-133 holds; see Remark S2.5.

Proposition 6.2. *For $k \geq 3$ let θ_k consist of two vertices x, y joined by k internally disjoint paths xa_ib_iy. Then θ_k is strong C_4-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

Proof. Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with j, i, l distinct) is induced. Conversely, let P be an induced path. If P avoids x and y, it lies inside one edge a_ib_i. If P contains x but not y, then x has at most two neighbours on P, each followed on P by at most its b-vertex, so $|P| \leq 5$. If P contains x and y, then every a_j on P is adjacent on P to x and every b_j to y, at most one branch lies entirely on P, and so P consists of x, y, at most one full branch and at most one further neighbour of each of x and y. Hence $|P| \leq 6$. □

Thus no argument with a uniform positive margin below 3 can prove GEN_3, already among graphs of radius 3. The graphs θ_k realise $(r, p) = (3, 6)$ with $\ell = 3 - 6/n < 4$. The proof of Theorem 5.9 is not a margin argument.

Let $\ell \leq 4$. Corollary 5.19 states that no connected graph in this range violates REAL-133. The case division is that of Proposition 5.16, and the remaining case $r = 3$, $p = 6$ and $S > 3n$ is excluded by Theorem 5.9. Proposition 5.16 records the weaker upper bound $4 - 16/n$ that Lemmas 5.14 and 5.15 gave for a hypothetical graph with $p \leq 6$ and $S > 3n$. That hypothesis does not occur. The range $\ell > 4$ remains open: Corollary 6.1 still leaves GEN_K for $K \geq 4$ and $\mathrm{RAD}(r)$ for $r \geq 5$.

7 | What is proved and what is open

The comparisons with Chen [3], Chen and Xu [5], Chen, Wu and Xu [4], Chudnovsky and Stacho [6], DeLaViña, Larson, Pepper and Waller [8], Karthick and Maffray [13], and Brandstädt and Hoàng [1] are in the introduction, in Section 5 and in Remarks 3.7 and 3.8. Table 1 records the results used for the range $\ell \leq 4$. An open row is not a hypothesis of a proved row. Outside Lemma 5.8 and Lemma 5.10, a finite search is not a proof here.
Lemma 5.14 gives $\ell < 4$ whenever $p \leq 6$. Theorem 5.9 replaces that estimate by $S \leq 3n$ throughout the range $p \leq 6$. This excludes the pairs $(r, p) = (2, 6)$ and $(3, 6)$ from the range $\ell \geq 4$, and from no wider range: the graphs θ_k realise $(3, 6)$ with $\ell = 3 - 6/n$. The same density restriction applies to induced-path order 7. If $p = 7$ and $\ell \geq 4$, then G contains a triangle. The condition does not forbid $(r, p) = (2, 7)$ in a triangle-free graph. The supplement gives a triangle-free example with parameters $(r, p, n, S, \ell) = (2, 7, 12, 34, 17/6)$, and it records the pairs that remain necessary in the bands $4 < \ell \leq 5$ and $5 < \ell \leq 6$.

Statement	Status
$M(n)$ for connected strong C_4-free P_6-free graphs, with uniqueness for $n \geq 7$	proved, Theorem 3.6
graphs with $r = 4$ and $p = 7$, the formula $S = 2n-b-m$, and the equality family	proved, Theorem 4.3 and Corollaries 4.4, 4.5
$2e \leq 3n$ for connected triangle-free strong C_4-free P_7-free graphs, equality only for the Petersen graph	proved, Theorem 5.4
the exact value $f(n)$ and the extremal graphs, $n \geq 10$	proved, Theorem 5.6
$S \leq 3n$ for connected strong C_4-free P_7-free graphs with at most one triangle	proved, Theorem 5.7
$S \leq 3n$ for every connected strong C_4-free P_7-free graph, equality only for the Petersen graph	proved, Theorem 5.9
$\Delta \geq 1$ for $n \geq 4$, hence $e \leq 2n - 4$ and $\ell < 4$; if $n \geq 6$ and $T \leq 2$, then $\Delta \geq 2$	proved, Lemmas 5.14 and 5.15; intermediate to Proposition 5.16
the cell $r = 3$, $p = 6$, $T \geq 2$, $n \geq 17$, $3 < \ell \leq 4 - 16/n$	intermediate estimate; empty by Theorem 5.9
REAL-133 for $\ell \leq 3$; for $r \leq 2$ and $\ell \leq 4$; for $r = 4$ and $\ell \leq 4$; for $r \geq 5$ and $\ell \leq 4$	proved, Theorem 5.18
REAL-133 for every connected graph with $\ell \leq 4$	proved, Corollary 5.19
REAL-133 for $\ell > 4$, and REAL-133 with no restriction on ℓ	open

TABLE 1 | Proved statements for $\ell \leq 4$, and what remains open.

The rooted bounds, the endpoint obstructions, the high-radius constructions, and the certificate for Lemma 5.8 are in the supplement. The conditional reduction of Theorem SB.1 is superseded by Theorem 5.9 and is not used in its proof. The proofs of Theorems 3.6, 4.3, 5.4 and 5.6 do not use the rooted bounds. The cut-vertex steps in Theorems 5.7 and 5.9 use Lemmas 5.3 and 2.3. No structure theorem for connected strong C_4-free graphs of radius $r \geq 5$ and path order $2r - 1$ is proved here.

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